

Indian Statistical Institute
Bachelor of Statistical Data Science
Year 2025, Semester 1
Mid-Semester Examination
Economics 1
Full Marks 30

Time 2 hours

Answer all the following questions.

1. (a) Explain the concept of opportunity cost.
(b) You were planning to spend Saturday working at your part-time job, but a friend asks you to go skiing. What is the true cost of going skiing? Now suppose you had been planning to spend the day studying at the library. What is the cost of going skiing in this case? Explain. (2+3)
2. Draw and explain the circular-flow diagram comprising household, firm, market for inputs and market for goods and services. Clearly mention the flows in real terms and in money terms. (5)
3. (a) Distinguish between movement along the demand curve and shift of the demand curve.
(b) Hardware and software for computers are complements. Discuss the effects on the equilibrium price and quantity:
(i) In the software market, when the price of computer hardware falls.
(ii) In the hardware market, when the price of computer software rises. (2+[1.5+1.5])
4. (a) Explain the concept of consumer surplus, producer surplus and social surplus using a suitable diagram without any kind of intervention in the market.
(b) How all the above surpluses are changed when the government imposes a price floor in the face of bumper production. (2+3)
5. (a) For the linear demand curve $p = 60 - 0.5q$, find the elasticity at $p = 10$.
(b) If the demand curve rotates outward with its vertical intercept unchanged (say to $p = 60 - 0.25q$), what happens to the elasticity at $p = 10$.
(c) If the demand curve shifts parallel to the right, what happens to the elasticity at $p = 10$. (2+2+1)

6. (a) Discuss the four properties of indifference curves.
- (b) Consider two Italian restaurants located in identical towns 200 kilometres apart. The restaurants are identical in every respect except for their tipping policies. At one restaurant, there is a flat \$15 service charge. At the other, a 15% tip is added to the bill. The average food bill at the first restaurant, exclusive of service charge, is \$100. How, if at all, do you expect the amount of food eaten in the two restaurants to differ.
- (2+3)
